

Demos con FPGAs Libres



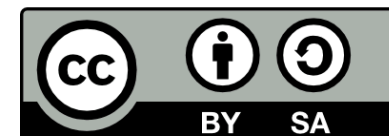
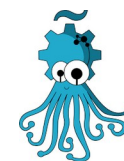
Juan González Gómez

@Obijuan_cube

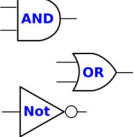
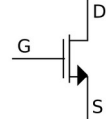
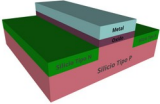
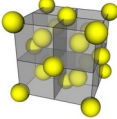
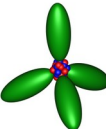
<https://github.com/Obijuan>



Museo Domus
11-Noviembre-2017
A Coruña



Situación

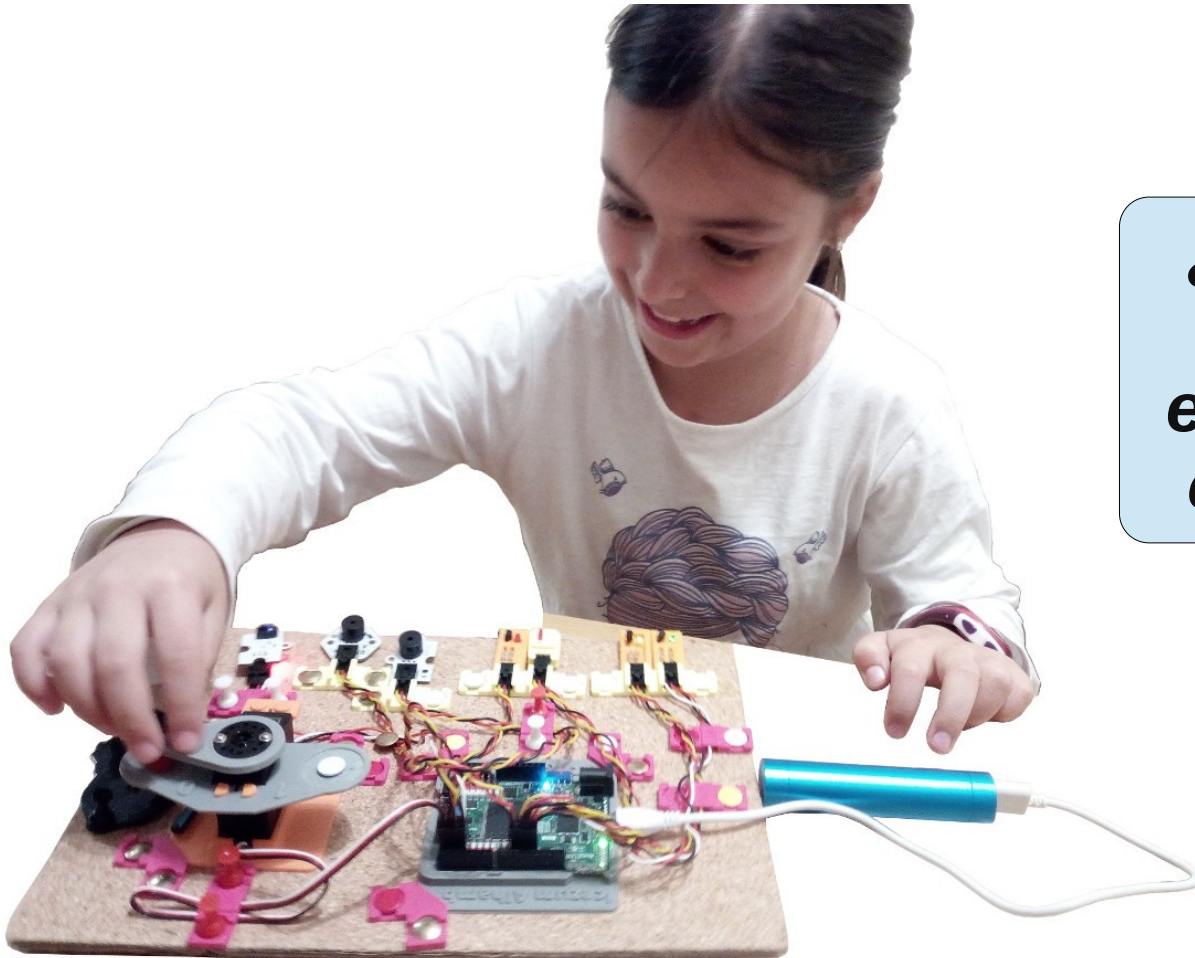
	Nivel 7: Software
	Nivel 6: Microprocesador
	Nivel 5: Puertas lógicas
	Nivel 4: Transistor
	Nivel 3: Semiconductores
	Nivel 2: Materiales
	Nivel 1: Átomos



← *Electrónica digital*

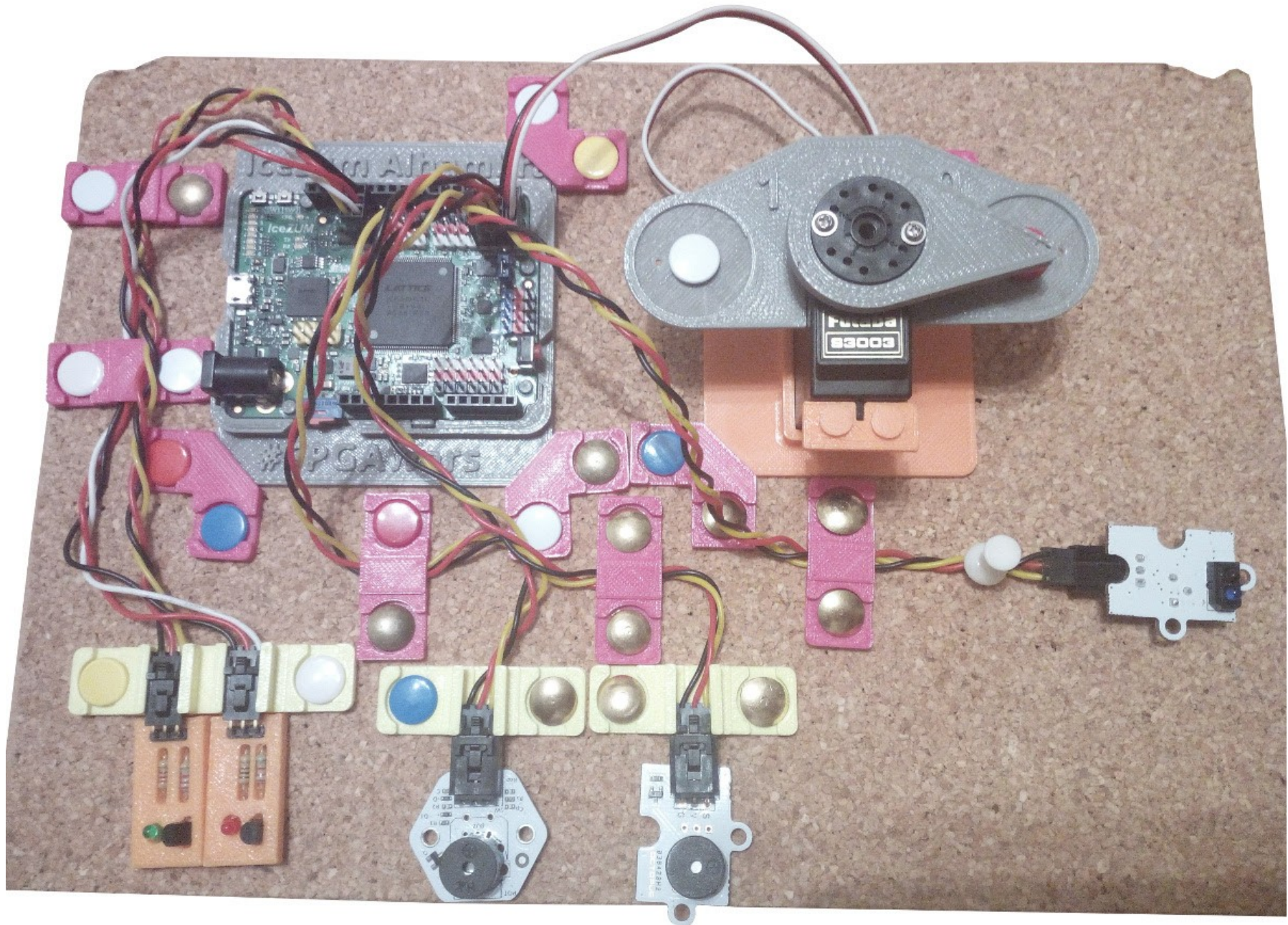
Motivación

Electrónica digital accesible



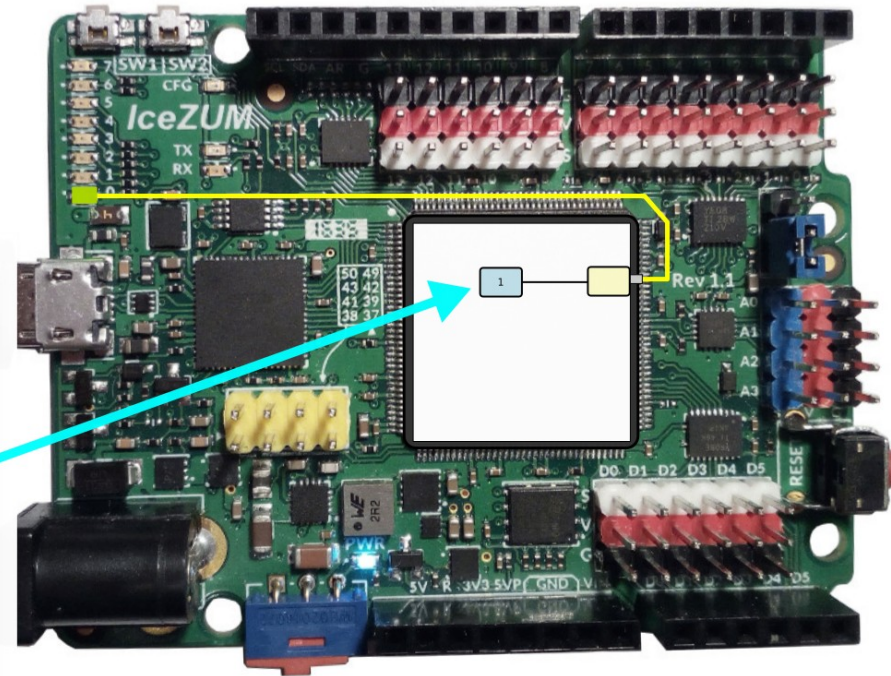
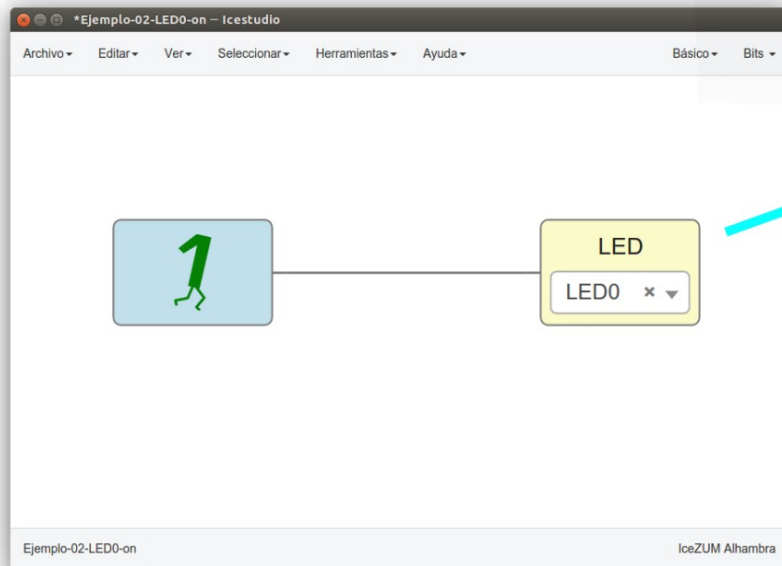
¿Cómo podrían los niños y los no electrónicos diseñar circuitos digitales?

¡Empezamos!

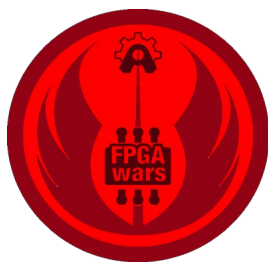
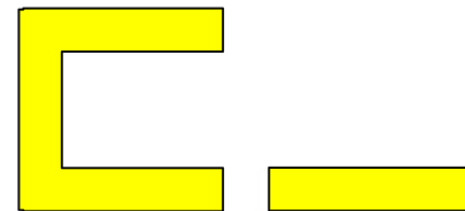
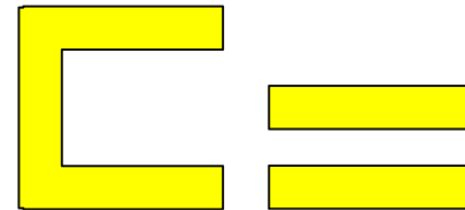
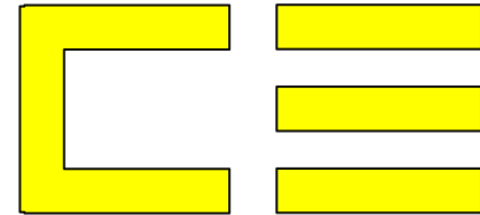
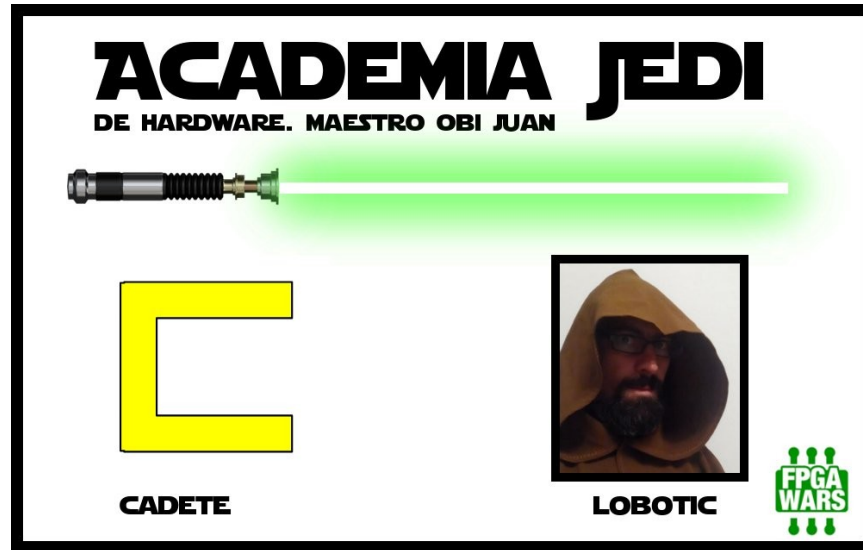


Encendiendo un led

ICESTUDIO

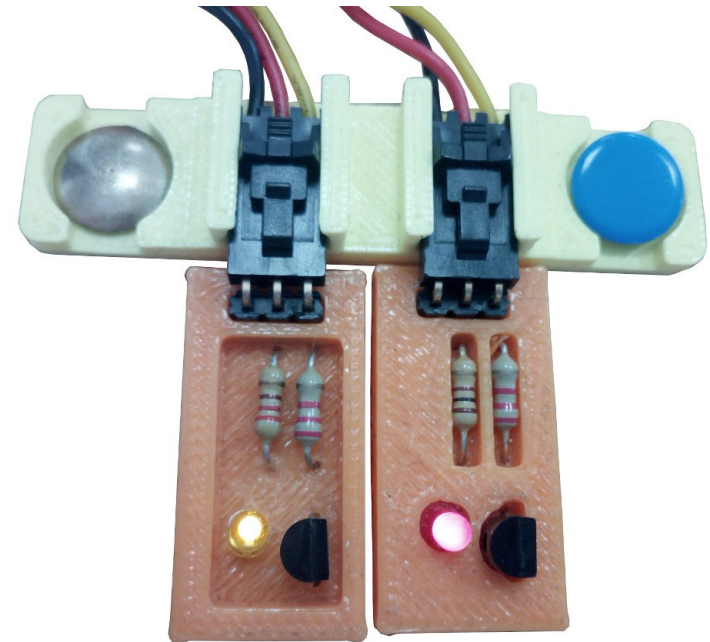
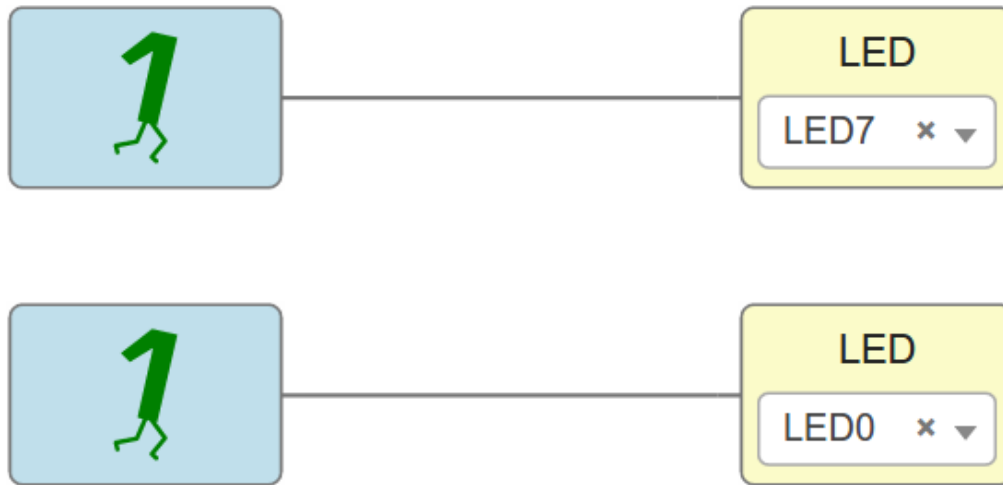


Tutorial de electrónica digital para makers con FPGAs libres

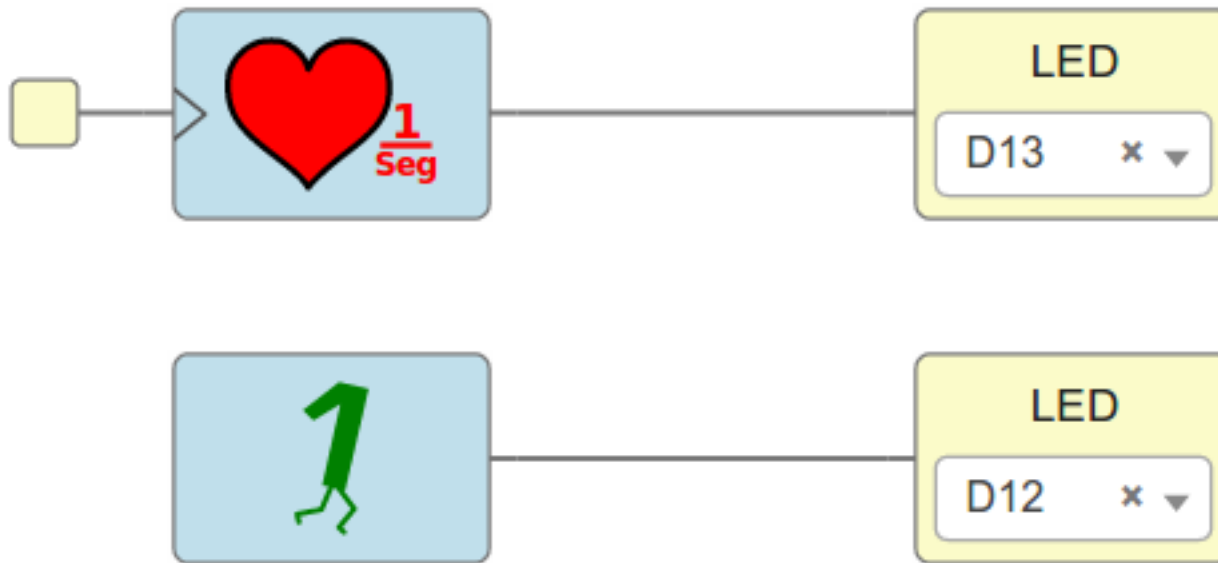


<https://github.com/Obijuan/digital-electronics-with-open-FPGAs-tutorial/wiki>

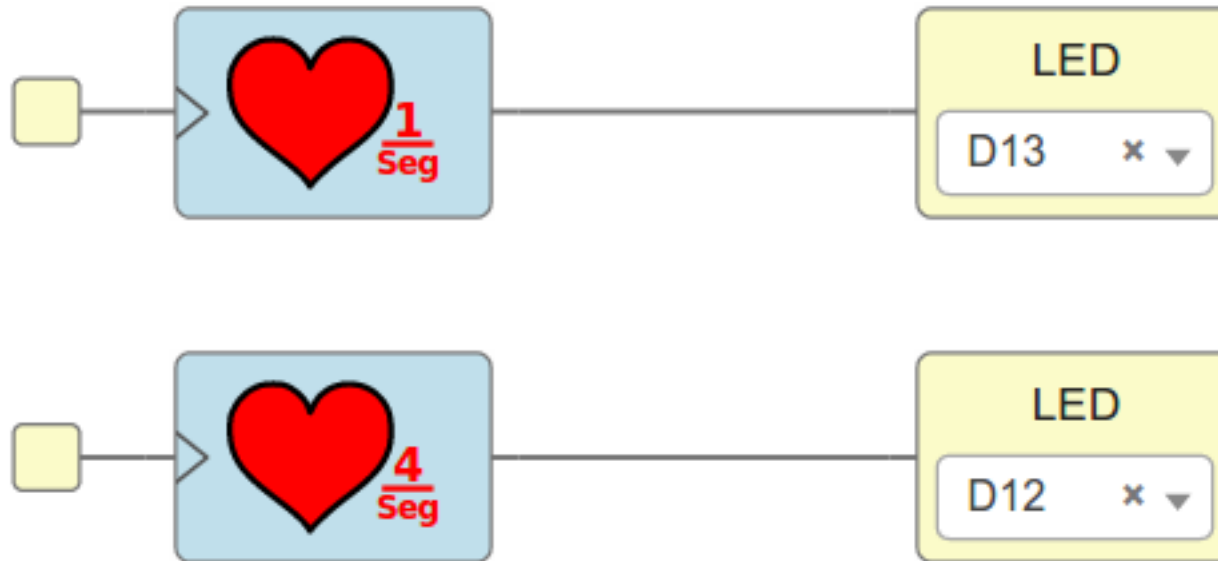
Circuitos en paralelo



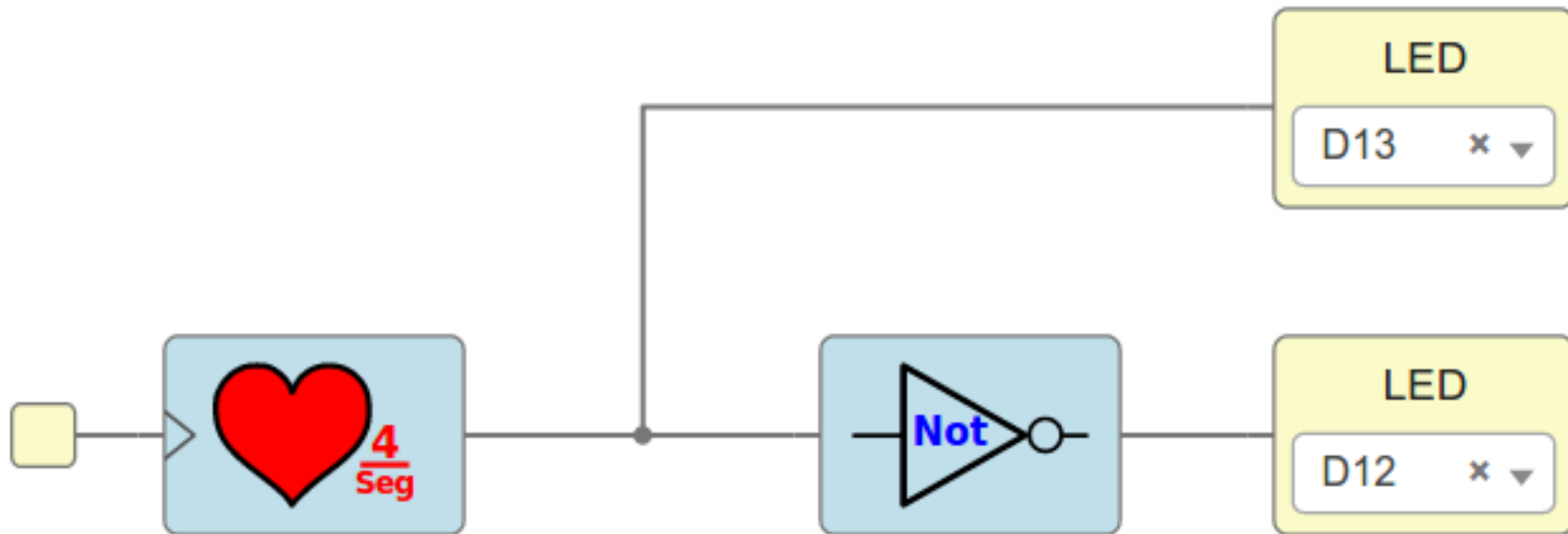
Bombeando bits



Dos leds parpadeando a la vez, a diferentes velocidades



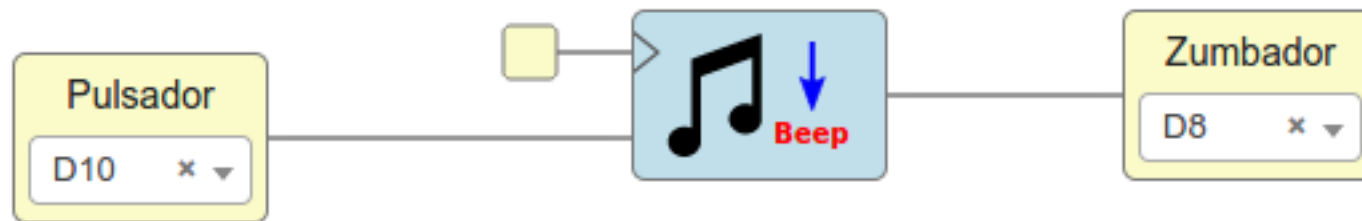
Sirena luminosa



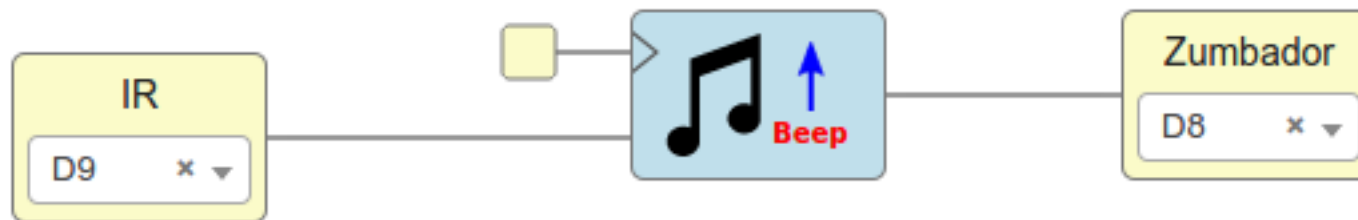
Juguemos a los supermercados



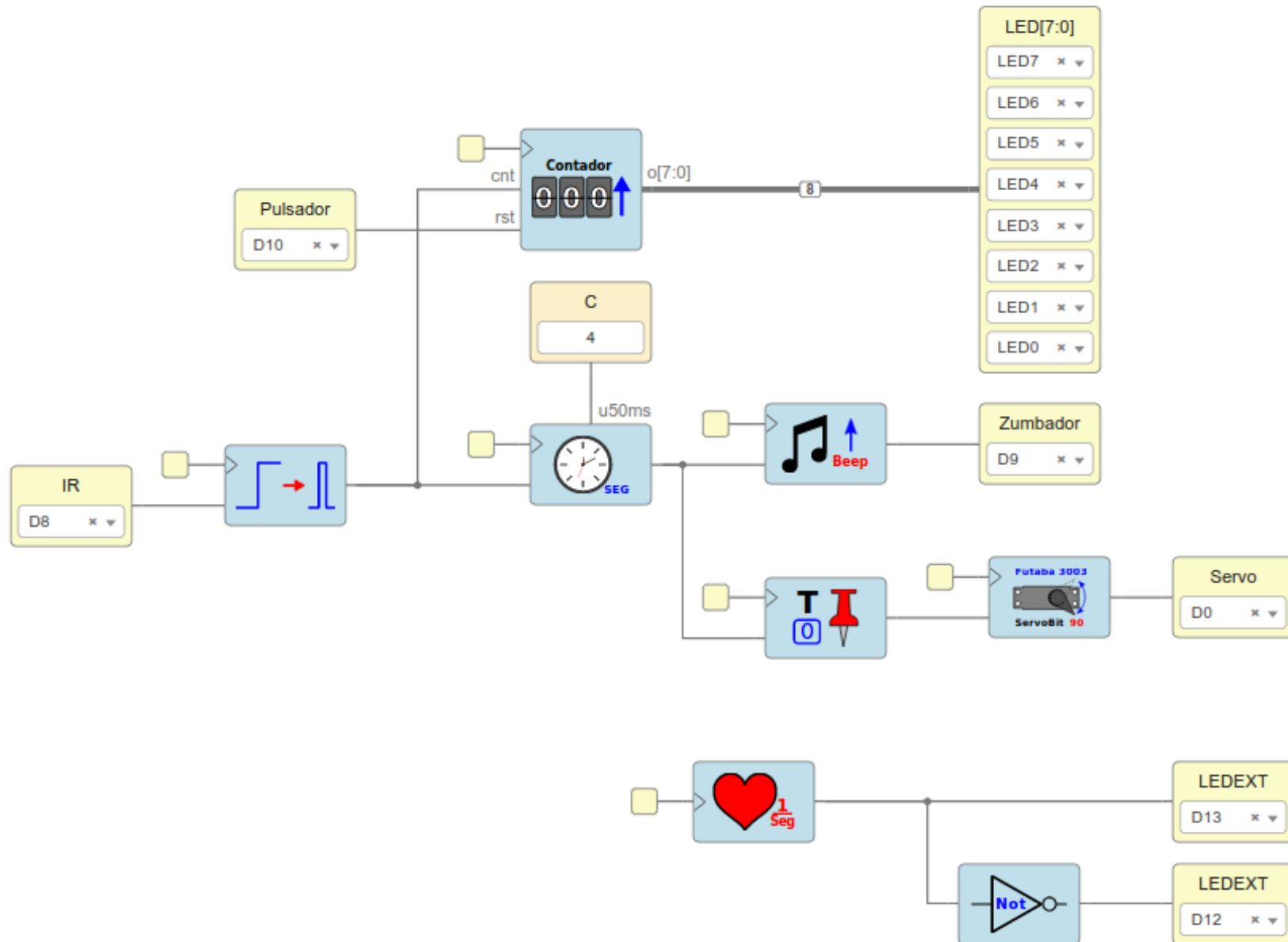
Timbre



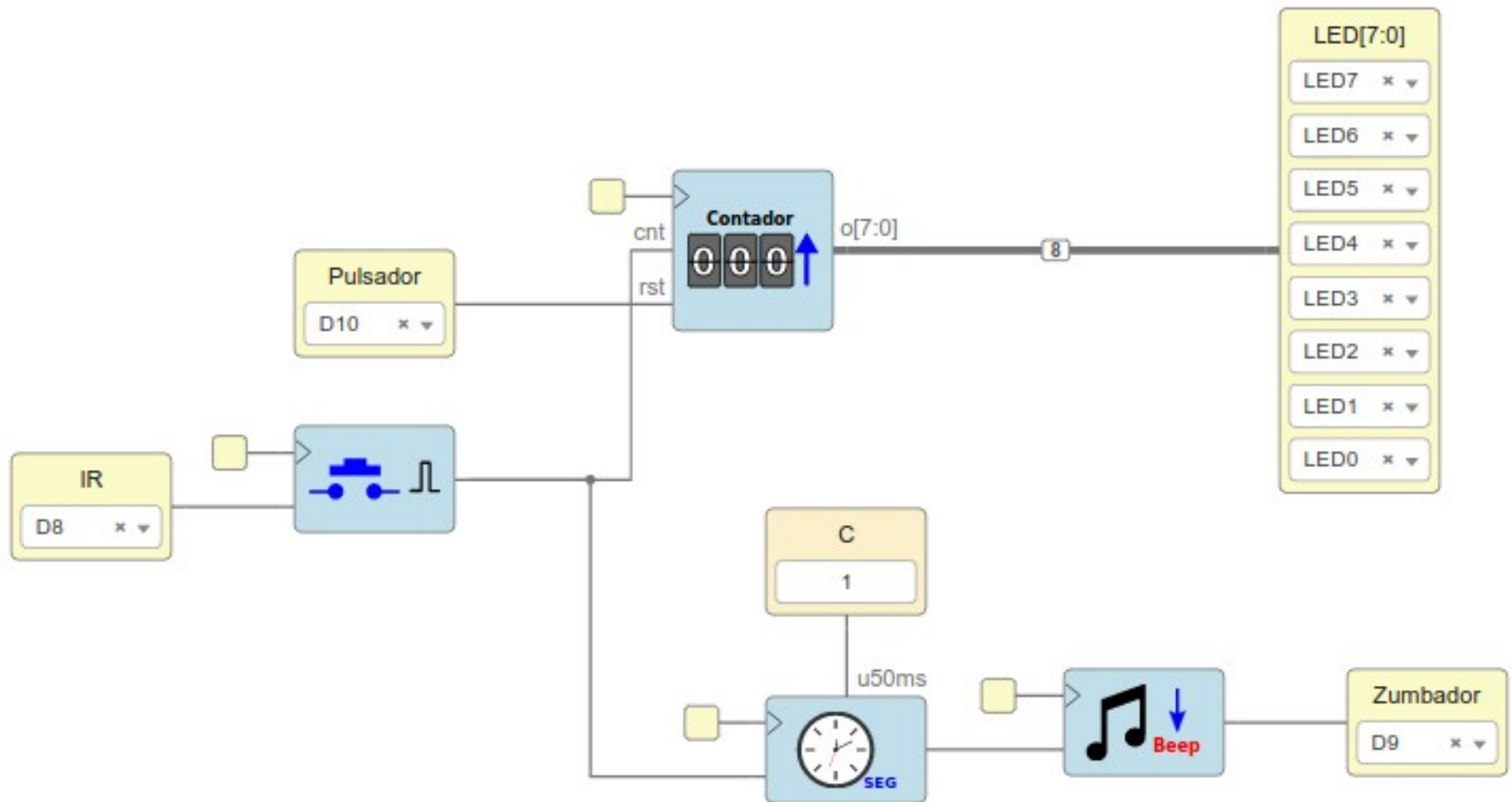
Escáner



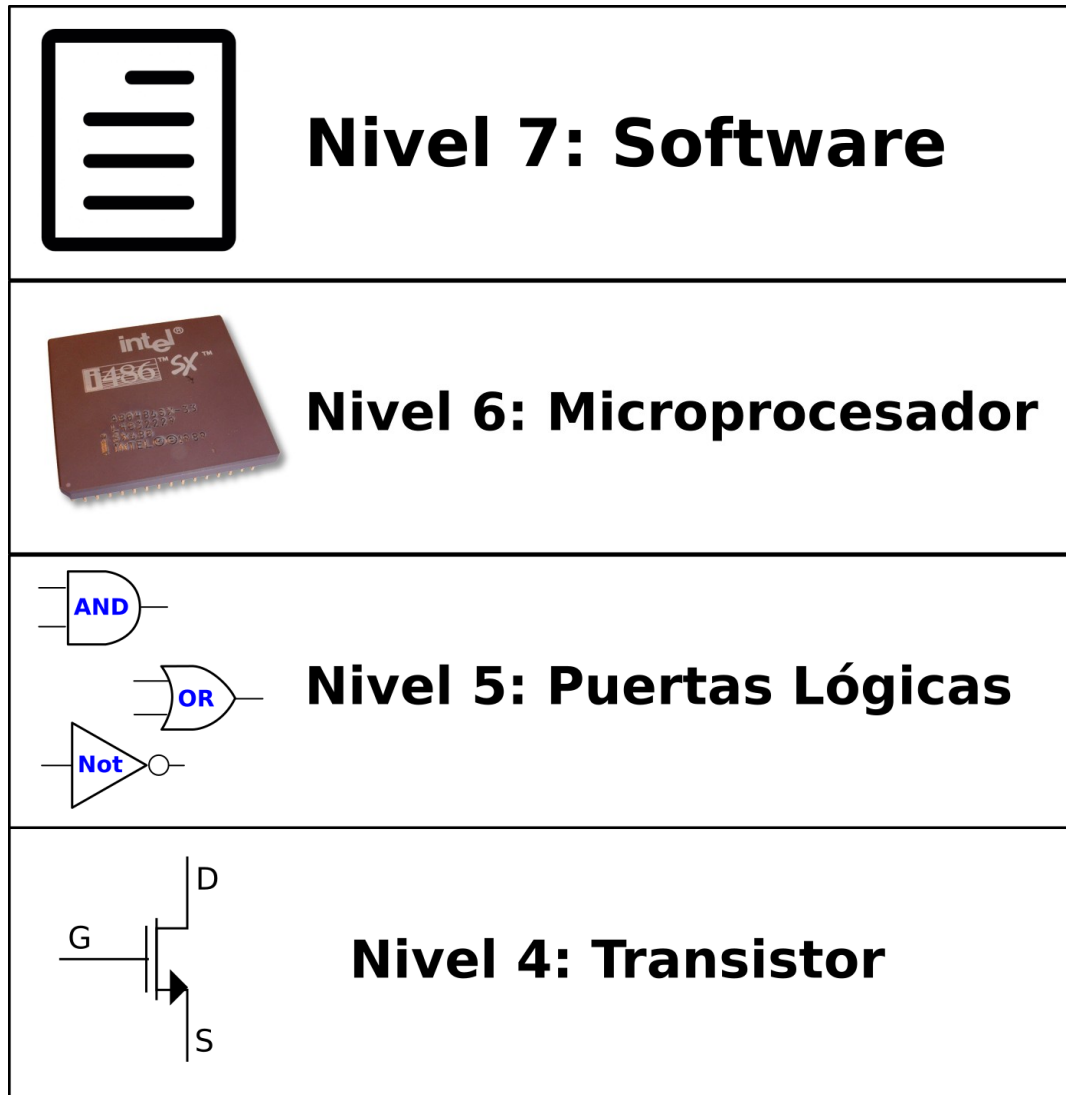
Escáner con contador y servo



Encoder con spiner



Lattuiño (II)



Lattuiño



FPGAs

Lattuino



https://github.com/INTI-CMNB/Lattuino_IP_Core

```
Lattuino_Counter | Arduino 1.8.2
File Edit Sketch Tools Help

Lattuino_Counter
// Lattuino Stick
// 4 bit counter

#define D1 14
#define D2 0
#define D3 1
#define D4 2
#define D5 3

#define DELAY 8

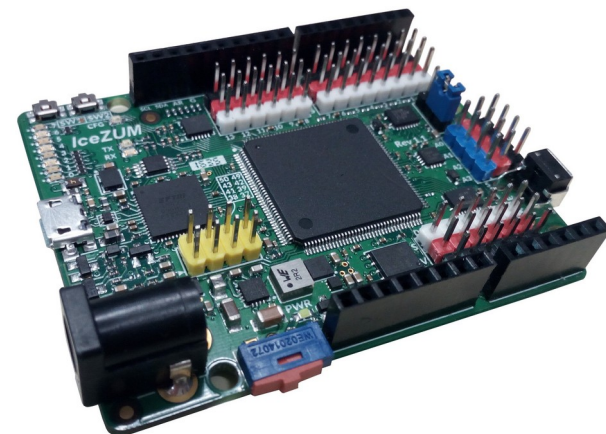
byte counter = 0;
int ledPin[] = {D4,D3,D2,D1};

Done uploading.

Sketch uses 496 bytes (35%) of program storage space. Maximum
Global variables use 10 bytes (7%) of dynamic memory, leaving

28 Lattuino Stick (2k) on /dev/ttyUSB1
```

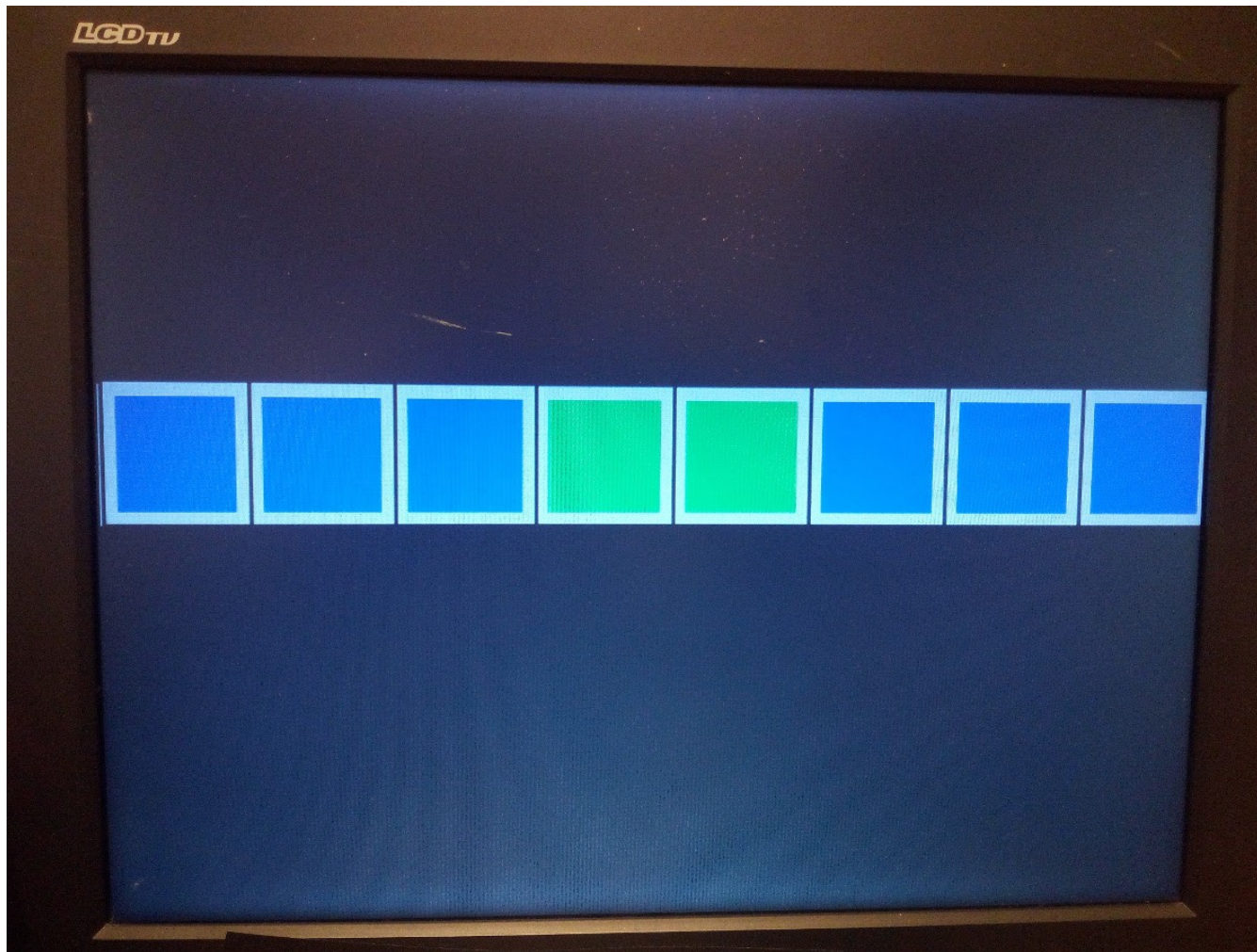
- Autor: **Salvador Tropea**
- Core de Arduino para FPGA
- Lattice Ice40 (1k, 4k, 8k)
- Migrando a Icestudio



VGA: Monsterled

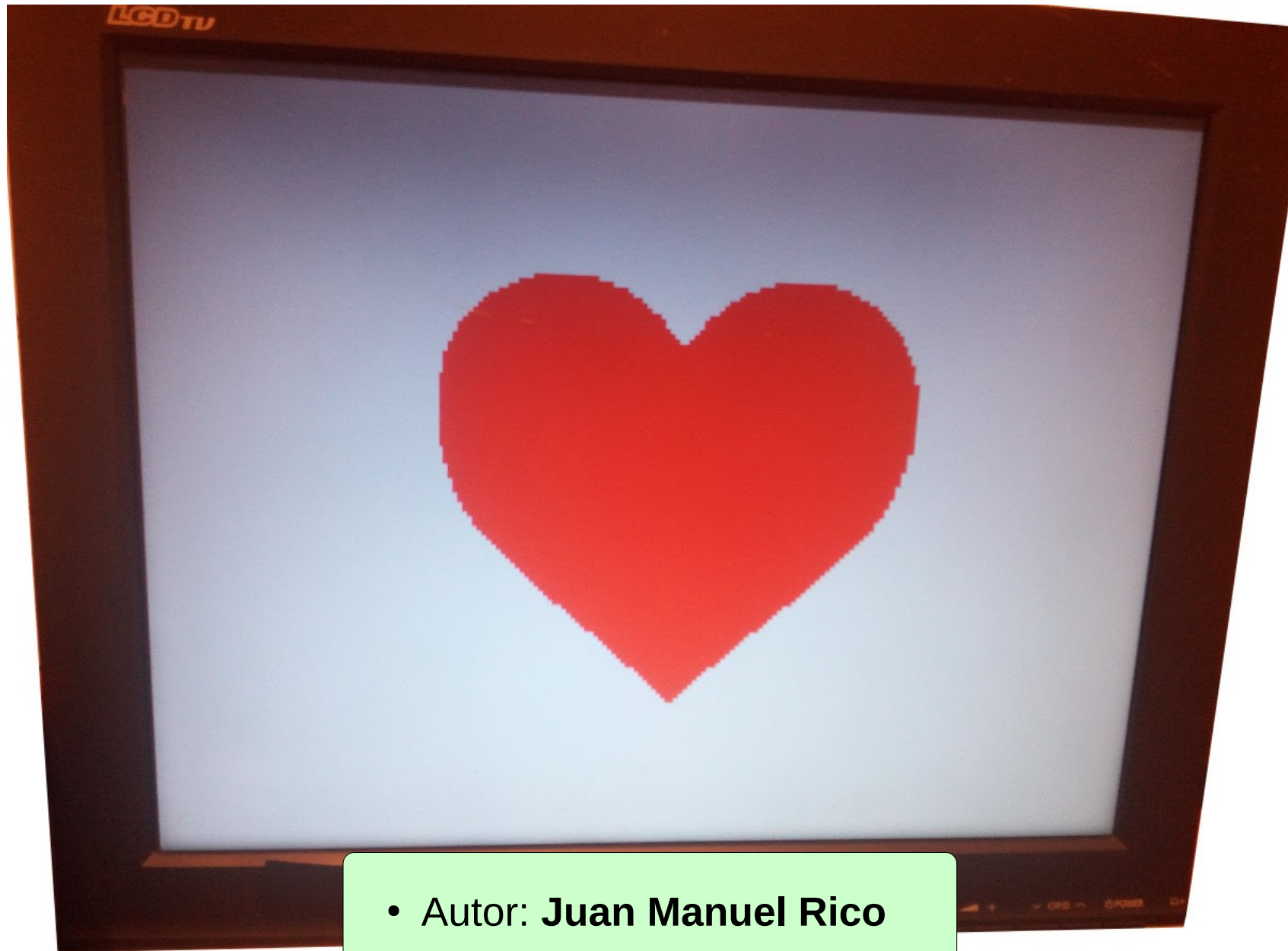


Screen-leds



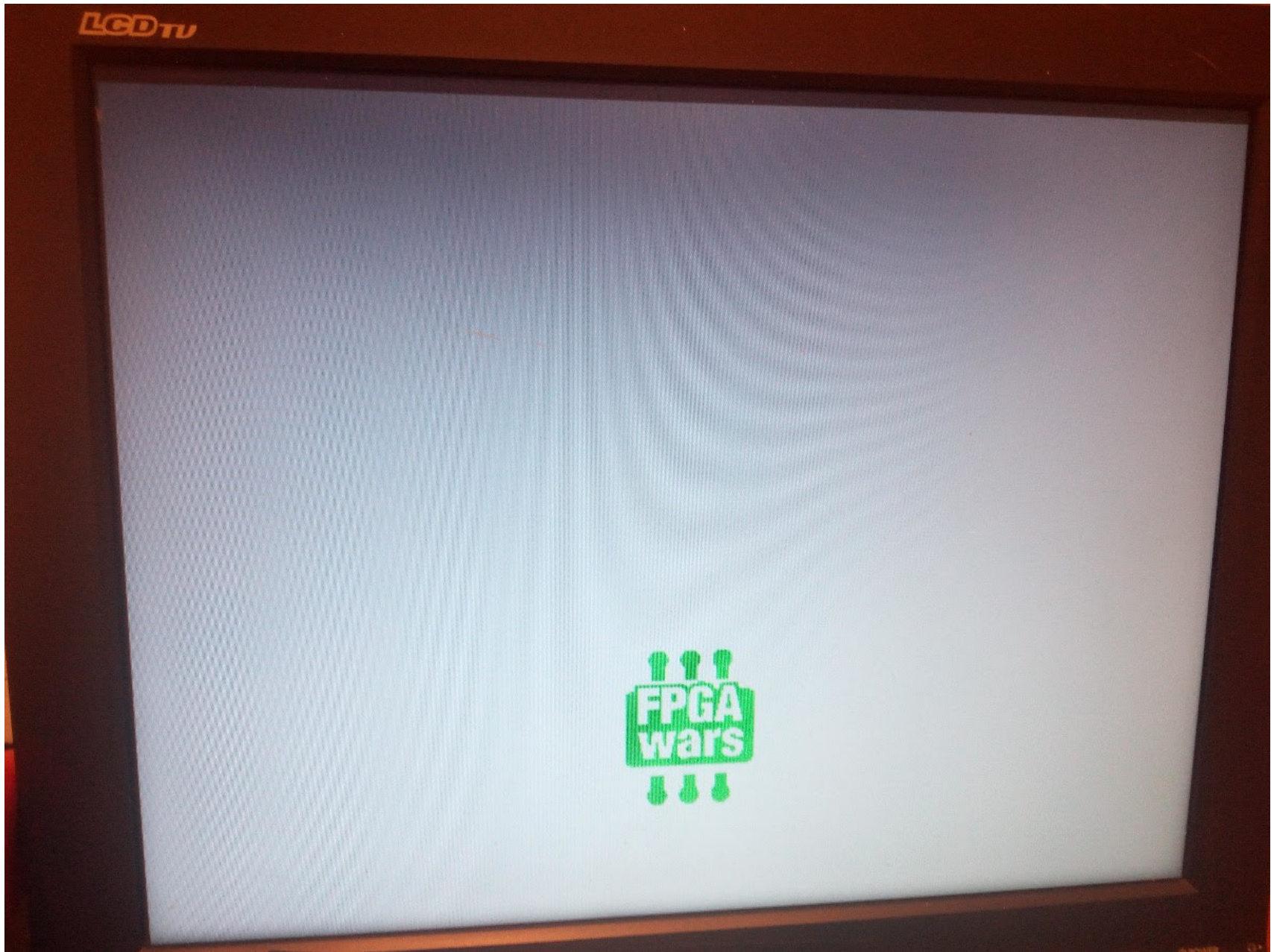
- Autor: **Juan Manuel Rico**

Screen-heart

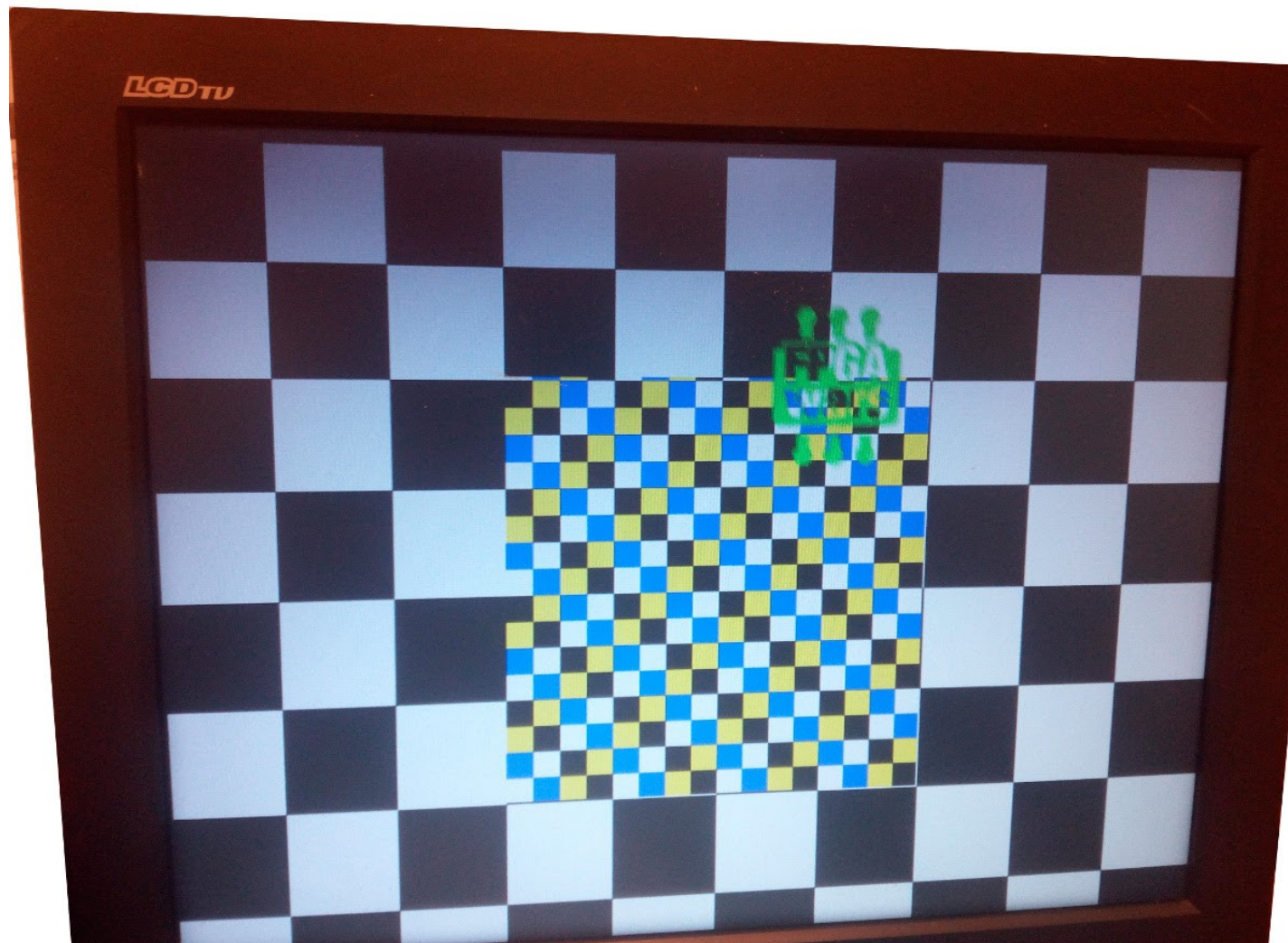


- Autor: **Juan Manuel Rico**

Screen-logo

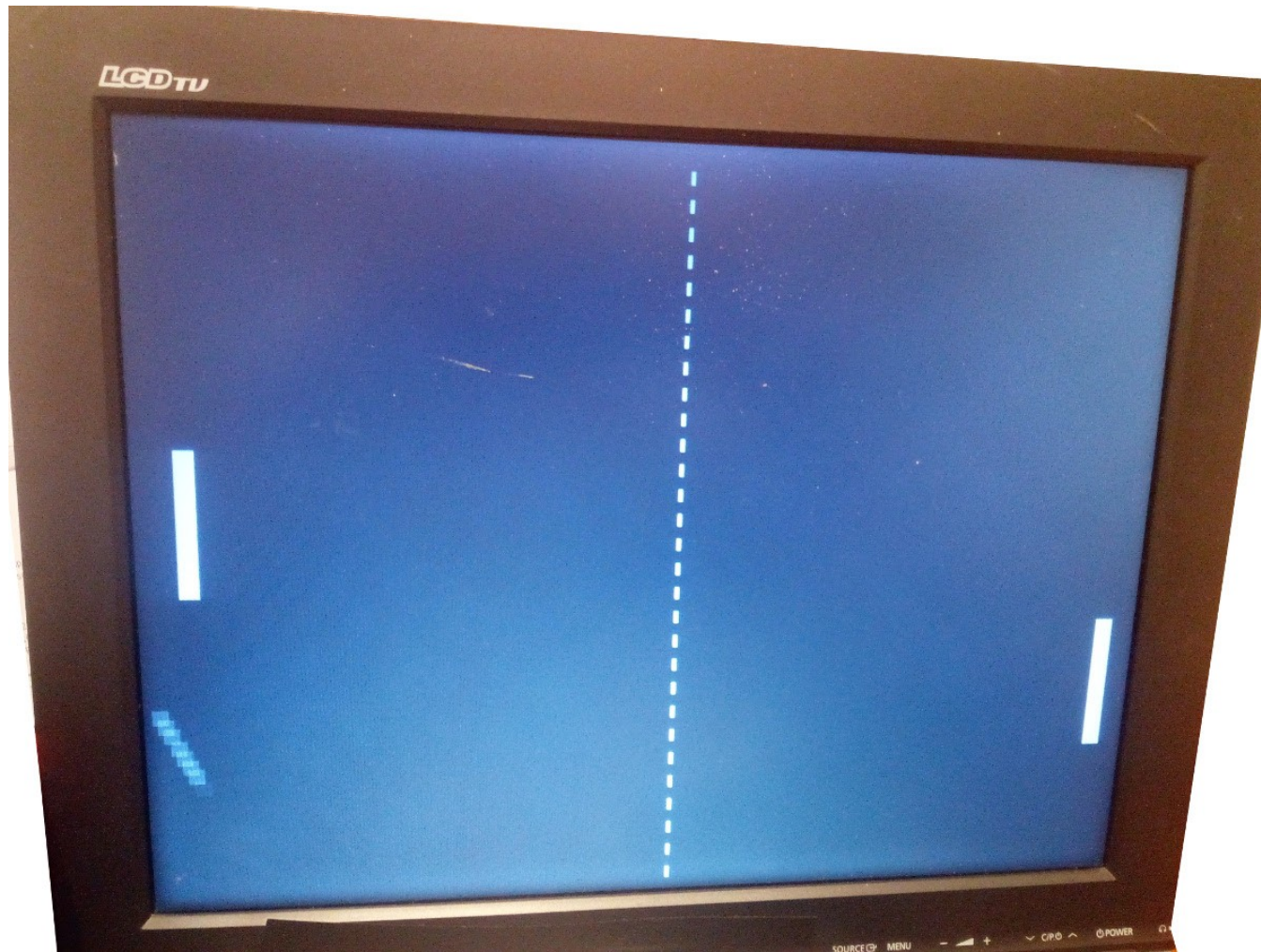


iPXs



- Autor: **Sergio Cuenca**

pong



- Autor: **Juan Manuel Rico**

**¡Que las FPGAs libres os
acompañen!**



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